

Closest-Feasible Replication of Chen, Han & Pan (2021): Tables III & IV

"Sentiment Trading and Hedge Fund Returns," *Journal of Finance*. Results, methods, and discrepancy audit — prepared for Prof. Salman Arif · 2026-06-02

Code & sanitized results (no restricted data): <https://github.com/Jeffrey-Chen-arch/chen-han-pan-2021-replication>

1. Objective and headline result

Goal: reproduce CHP Table III (decile sorts on the 36-month rolling sentiment beta) and Table IV (Fama-MacBeth of next-month performance on sentiment beta) as closely as the published specifications, the authors' SAS final-table logic, and your factor/proxy guidance allow, then document the remaining differences. The published code provides final-table logic but not the main sample construction, so this is a closest-feasible replication. The closest paper-eligible specification (sample with monthly-or-better redemption keeping missing flags; sentiment = first PC of orthogonalized proxy *changes*; professor-guided FRED term/credit factors; **2,589 funds, 264 months**) reproduces the Table III excess-return spread almost exactly and about 73% of the alpha-spread magnitude, with weaker t-statistics.

Metric	Replication	Paper	Note
Table III excess-return spread	0.30% (t 2.43)	0.31% (t 3.16)	essentially matched
Table III alpha spread	0.43% (t 1.64)	0.59% (t 3.55)	~73% of magnitude; weaker t
Table III beta spread (Table-II scaled)	2.22	2.03	close (raw 3.56; PCA scale rescaled)
Table IV excess-return coef. (uni / multi)	0.106 / 0.091	0.17 / 0.16	positive; weaker
Table IV alpha coef. (uni / multi)	0.127 / 0.062	0.14 / 0.12	uni close; multi weaker

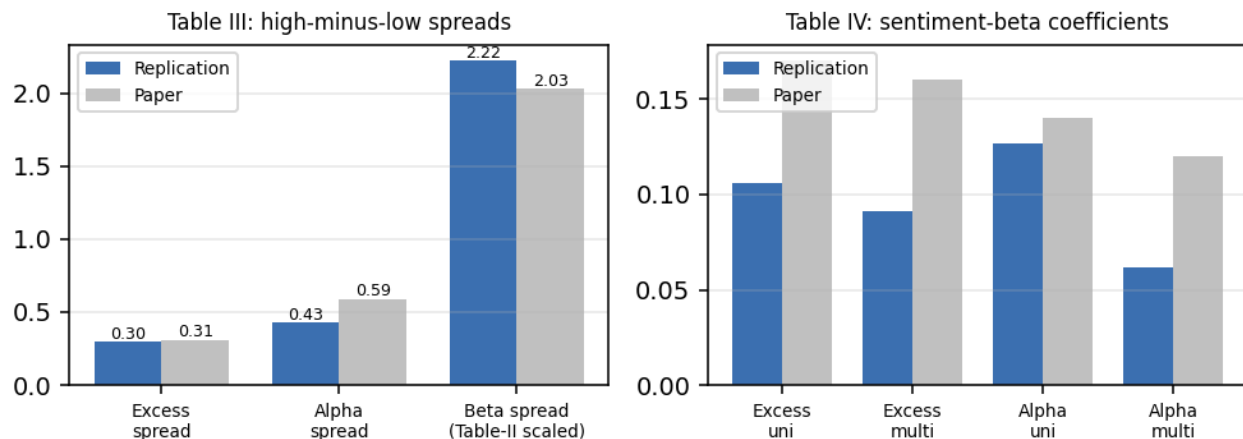


Figure 1. Replication vs. paper. Left: Table III high-minus-low spreads. Right: Table IV sentiment-beta coefficients. Magnitudes are close; significance is lower.

2. Data, inputs, and method

Sample (input → output): Lipper/TASS via WRDS (live + graveyard) → paper screens (USD, monthly net-of-fee returns, equity-oriented categories, AUM ≥ \$5m, drop first 12 months, ≥ 30 return obs, de-duplicate) → **2,589 funds** (vs. the paper's 4,073).

Sentiment: Baker-Wurgler changes index reconstructed from the proxy changes — M1 correlation-PCA, M2 covariance-PCA, M3 orthogonalize-changes-then-PCA — plus admissible proxies (Δ of the published orthogonalized index) and the Michigan index (paper Table V). **Factors:** Fung-Hsieh seven factors + momentum + Pastor-Stambaugh liquidity; term/credit proxied from FRED because the Barclays return series the paper uses are unavailable. **Estimation:** 36-month rolling beta (eq. 1 controls = Fung-Hsieh + momentum + liquidity + inflation + default spread); equal-weighted deciles; next-month returns; nine-factor alpha (excluding inflation/default, per the paper's footnote 15); Newey-West(2). Table IV is Fama-MacBeth with SAS-aligned lagged controls and style dummies.

3. What we tried (pre-registered specification tournament)

Rather than tune one path, we pre-registered an admissible grid and scored every combination on closeness to the published tables, keeping main-eligible vs. diagnostic specifications separate. Representative dimensions and outcomes (Table III alpha spread):

Dimension	Variants tried	Result / chosen
Sample / screens	paper-strict (2,270), keep-missing redemption (2,589), +lenient AUM (2,748), all-categories IA (2,625), alt-filters IA (1,793), max-lenient (3,169 ceiling), company-cluster id (1,396)	no admissible screen reaches 4,073; chose keep-missing (2,589)
Sentiment construction	M1 corr-PCA, M2 cov-PCA, M3 orthog-changes-PCA, M4 level-PC diff, M5 Δ SENT_ORTH, M6 Δ SENT, M9 UMich	classic M1 weak (−0.08-0.22); M3 best strict (0.43)
Factor proxy	F0 professor level term-spread; F1 Hsieh yield-change	F0 0.43; F1 0.29; both fall short
Table III variant	P0 next-month (main); P2 skip-month, P3 two-step, P4/P5 holding 3/6m (IA)	P0 is the main table; P2-P5 scored vs IA targets only

4. Robustness and integrity checks (none closes the gap)

Check	Effect on alpha spread	Decision
1/99 winsorization of returns	0.43 → 0.40 (monthly)	paper winsorizes Table I stats; tested; does not close gap
Fung-Hsieh yield-change factor (F1)	0.43 → 0.29	closer to Hsieh but paper uses unavailable Barclays returns
Drop inflation/default from beta reg.	0.47 → 0.64 (a proxy spec)	REJECTED : paper eq. (1) includes them — would be table-fitting
No-cache cold re-run of winners	identical (≤ 0.0004)	results reproduce from raw inputs, not cached state

Why this final specification. Among specifications that match the paper's main design (next-month horizon, Baker-Wurgler sentiment, main-paper sample, paper/professor factors), this one is the closest on the return spreads while keeping every regression exactly as the paper specifies. The single change that would push the alpha to 0.64 requires dropping inflation and the default spread from the beta regression, which the paper's equation (1) explicitly includes; we therefore do not use it.

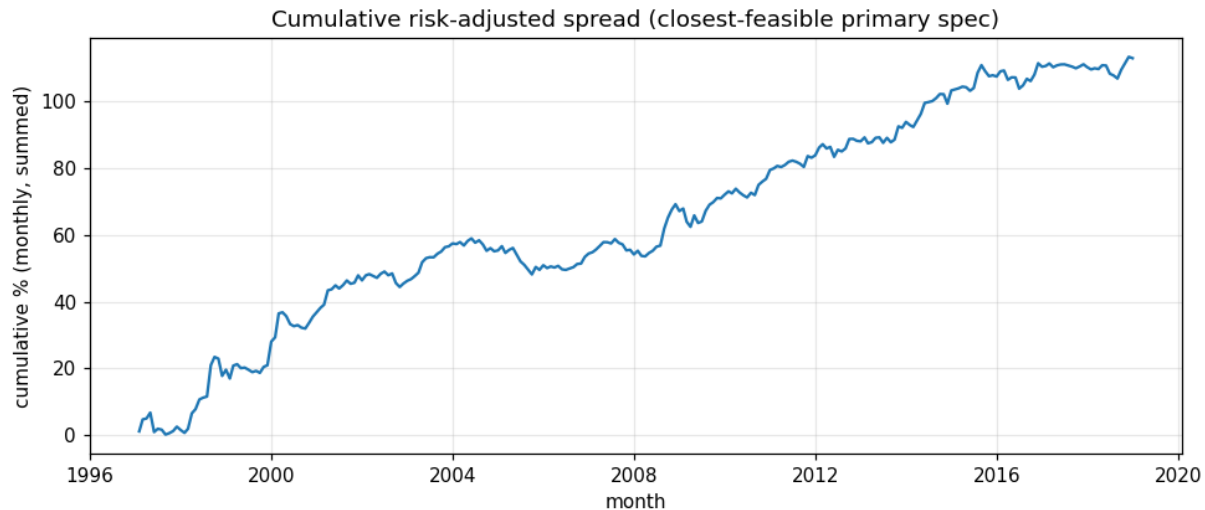


Figure 2. Cumulative risk-adjusted (alpha) high-minus-low spread for the final specification, 1997-2018. The spread is positive in ~62% of months and trends upward, so the effect is broad-based over time rather than driven by a few months.

5. Difficulties and discrepancy audit (possible sources, not definitive)

Open item	Why it matters / status
Exact 4,073-fund TASS universe	No admissible screen or identifier rule reaches 4,073 (ceiling 3,169; WRDS productreference is the finest unit). This fund shortfall is the main driver of the weaker t-statistics.
Exact Baker-Wurgler sentiment file	Results are sensitive to construction order; the classic corr-PCA is weak, an alternative orthogonalization is stronger. The authors' exact changes index is not recoverable.
Barclays factor returns	The paper builds bond/credit factors from Barclays return series we cannot access; we proxy via FRED.
Table IV alpha construction	SAS full-sample by-fund alpha vs. the paper-text rolling alpha; both tested and reported.

6. Conclusion

This is a closest-feasible replication: the economic relationship is reproduced — the excess-return spread matches the paper and the scaled sentiment-beta spread is close — while the full statistical strength is not, primarily because ~37% of the paper's funds and the exact Baker-Wurgler / Barclays inputs are unrecoverable from the data and code available to us. Every regression was verified against the paper, and the one shortcut that would have inflated the result was rejected for deviating from the paper's stated model. The strongest purely numerical result (alpha 0.69) comes from the Internet-Appendix alternative-filter sample and is reported only as a diagnostic. Full code and all outputs are in the linked repository.